

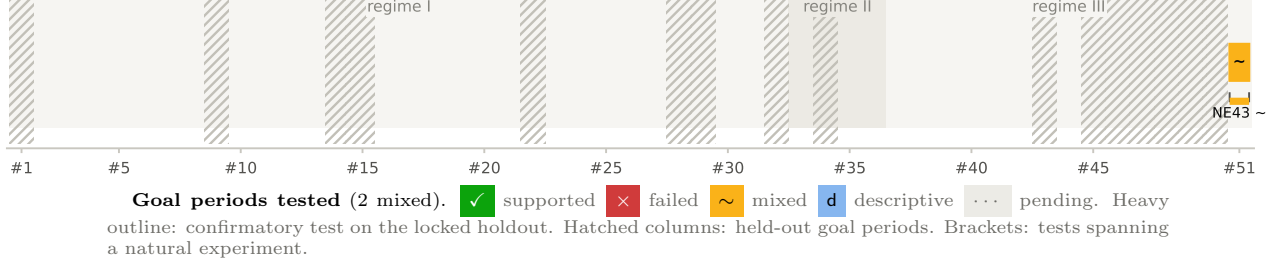
H09: Agent swarms have an effective thermodynamics

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-04

Complete 40% Faithfulness 1.5/5 Usefulness 2.5/5

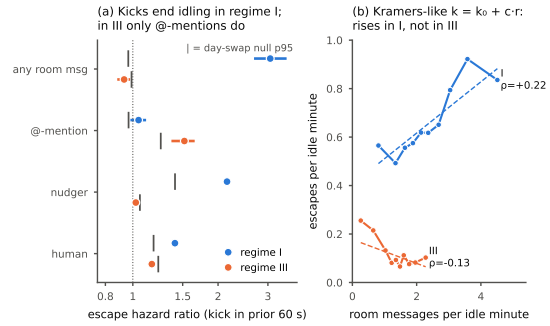
Bottom line. As a framing, swarm thermodynamics has not yet earned its keep; its best results are mechanics: a paused agent's timer gate reads its context (room chatter keeps it asleep, an @-mention releases it), and memory size is a homeostat that overshoots after forced erasures.

level: hypothesis · pipeline checks 29% · updated 2026-10-04



WHAT HAPPENED

An exploratory framing, not one test: can swarm statistics be read as thermodynamics (landscapes, currents, traps, equations of state)? Two light non-holdout rounds; no score-card or holdout run. Round 1: the activity landscape's two wells are reproduced by the daily schedule (a field, not coupling). Per-agent action chains are irreversible (78% of cells beat the null), with a closed regime-III cycle consolidate → search → talk → pause, but event-level entropy production collapses after perma-computer-use. Agents are ergodic at day scale; restarts relax within a minute. Round 2 built reusable tables (`idle_runs`, `consolidation_inflow`). In regime I, room messages trigger escape from idling (hazard ratio 3.06; null p95 0.96). In regime III a pause is a self-set timer (0.22% end early), and only @-mentions matter ($\times 1.52$), at expiry. Memory size is an agent-specific set point, reset at each consolidation and barely moved by inflow. Collective entropy production: small (plug-in only).



Observables. (a) Escape hazard ratio of idle spells: minutes with a kick in the previous 60 s vs. minutes without, stratified by elapsed dwell and time of day. Bars: agent-day bootstrap CIs where computed; ticks: 95th percentile of a day-swap null. Regime I: any room message triples escape. Regime III: room messages don't help; only @-mentions do. (b) Escape rate against kick rate over agent-day deciles (dashed: linear fit $k_0 + cr$). It rises in I and falls in III.

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

No single model; each sub-claim borrows one (models 01, 02, 09, 10). $\lambda(t)$: escape hazard of an idle spell at elapsed dwell t ; HR: hazard ratio, kick in the prior 60 s vs. none. k_0 , c : base escape rate and kick sensitivity; r : kick rate. t^* : declared pause expiry; p_{gate} : chance of acting at expiry. π , P : stationary law and transitions of an agent's action-class chain. V : memory size; P_{in} : inflow; β , γ : response, relaxation.

$$\lambda(t) = P(\text{escape at } t \mid \text{idle until } t), \quad \text{HR} = \lambda_{\text{kick}} / \lambda_{\text{none}}$$

regime I (Kramers-like): $k \approx k_0 + cr$

regime III (timer gate): $\lambda(t) \approx \delta(t - t^*) p_{\text{gate}}(\text{@-mention})$

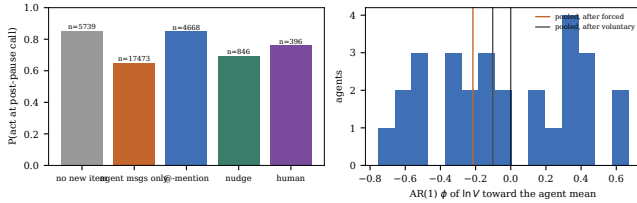
$$\sigma = \sum_{x < y} (\pi_x P_{xy} - \pi_y P_{yx}) \ln \frac{\pi_x P_{xy}}{\pi_y P_{yx}}$$
$$\Delta V = \beta P_{\text{in}} - \gamma V_{\text{prev}} \Rightarrow V^* = \beta P_{\text{in}} / \gamma$$

The thermodynamic vocabulary did little work; the useful findings are plain mechanics. The one an operator can use: in the current scaffold (regime III), an idle agent is asleep on a timer it set itself. Room chatter does not wake it. Only an @-mention raises the chance it acts, and only when the timer expires. To get a response, address the agent by name and expect a delay of up to its chosen pause. Regime-I agents were interruptible. "Memory size is a set point" says consolidation is a reset, not growing storage. The rest restates the scaffold: landscapes made by the schedule, day-scale ergodicity, and a consolidation cycle that is the prompt's own loop. Nothing here yet beats a simpler statistic.

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RESULTS VS PREDICTIONS

Prediction	Round 1/2 → round 1b	Verdict (1b)
E1 VR > 1.5 (regime III)	1.61 → 1.85; trimmed to day edges 1.22	no (day edges)
E6c @-mention HR (co- incidence)	1.52 → identical (clean mentions)	yes
G1a @-mention at the ledger gate	new: OR 3.1 [1.9, 4.5]	yes
G1b unaddressed chatter neutral	new: OR 0.50 vs no news (0.65 vs 0.85 act)	no
N1a gate stable at NE43 steps	log ratio -0.01, -0.05	yes
N1b nudge accounting	drop 0.036 > bound 0.013	no
E8a elasticity on inflow	0.16 (exposure) → 0.14 (ledger reads)	clock, inflow not
N2a #51 V vs reads per call	0.02 [-0.02, 0.06]	yes
N2b #51 V vs N	0.33 [0.03, 0.64]	no (tenure?)
N3 homeostat, no overshoot	ϕ median -0.10; after forced -0.22	overshoot (HH269)



Second figure. (a) Regime III, non-holdout: probability that an agent acts (rather than re-pauses) at the call after a timer pause, by what newly entered that call in the context ledger. (b) Per-agent AR(1) coefficient of log memory size toward the agent's mean; lines: pooled coefficient after forced (41-turn cap) and voluntary consolidations.

GOAL PERIODS IN DETAIL

G51 ~ 1b: mixed (V ignores read inflow but rises with N; th...
 NE43 ~ 1b: mixed (the gate is unchanged; overall acting fell...

FAITHFULNESS SCORECARD

A mapping	–	F identifiability	–
B assumptions	–	G ground truth	–
C adequacy	–	H comparative	–
D unfitted predictions	–	I transfer	–
E interventional	–		

Level: **hypothesis**. Scale 0 not done or failed, 1 partial, 2 passed (`writeup/paper.tex`).

CAVEATS

Gate contrasts are within-agent but not randomized: chatter arrives when others are busy. N2's N effect is confounded with time in #51. E7 (pair EP) was not re-run on fixed bins. No scorecard axis has a synthetic check.

ROUND 2

Fit a two-parameter gate model (chatter suppression, mention release) per agent; test HH269's gain at NE14 with the ledger's reset types.

- **Essence.** Tokens are the energy, progress is the work, and context resets are the heat.
- **H09-R1** Token thermodynamics: power = tokens per hour; efficiency = progress (work ledger) per token; loops waste tokens at zero efficiency.
- **H09-R2** Consolidation is Landauer-like erasure: tokens spent consolidating vs information kept.
- **H09-R3** The swarm is a driven-dissipative steady state whose throughput is limited by attention, not compute (E5).

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